数列专题习题册（中英双语版）答案与解析

一、选择题（I. Multiple Choice Questions）

题号	答案	解析（Explanation）	英文解析（English Explanation）
1	A	等差中项公式:若 a,A,b 成等差数列，则 $A=\frac{a+b}{2}$。题目中两数为 $\sqrt{5}$ 和 $\sqrt{5}$，代入公式得:$A=\sqrt{5}$，选 A。	Arithmetic mean formula: If a,A,b are in arithmetic sequence, then $A=\frac{a+b}{2}$.The two numbers in the question are $\sqrt{5}$ and $\sqrt{5}$. Substituting into the formula: $A=\sqrt{5}$, so choose A.
2	B	等差数列通项公式:$a_n=a_1+(n-1)d$。	Arithmetic sequence general term formula: $a_n=a_1+(n-1)d$.
3	C	等比中项公式:若 a,G,b 成等比数列，则 $G=\hat{A}\pm\sqrt{ab}$。代入 $a=4$，$b=16$，得 $G=\hat{A}\pm\sqrt{4\times16}=\hat{A}\pm8$，选 C。	Geometric mean formula: If a,G,b are in geometric sequence, then $G=\hat{A}\pm\sqrt{ab}$.Substitute $a=4$ and $b=16$, we get $G=\hat{A}\pm\sqrt{4\times16}=\hat{A}\pm8$, so choose C.
4	D	等比数列公比公式:$q=\frac{a_{n+1}}{a_n}$。已知 $a_2=6$，$a_3=12$，则 $q=\frac{12}{6}=2$，选 D。	Geometric sequence common ratio formula: $q=\frac{a_{n+1}}{a_n}$.Given $a_2=6$ and $a_3=12$, then $q=\frac{12}{6}=2$,

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			so choose D.
5	A	等差数列通项公式:$a_n = a_1 + (n-1)d$。已知 $a_1 = 3, d = 2, n = 5$, 则 $a_5 = 3 + 4 \times 2 = 11$, 选 A。	Arithmetic sequence general term formula: $a_n = a_1 + (n-1)d$ .Given $a_1 = 3$, $d = 2$, $n = 5$, then $a_5 = 3 + 4 \times 2 = 11$, so choose A.
6	A	等差数列前 n 项和公式:$S_n = \frac{n(a_1+a_n)}{2}$。已知 $a_3 = 7$，$S_3 = 15$，则 $15 = \frac{3(a_1+7)}{2}$，解得 $a_1 = 3$。由通项公式 $a_3 = a_1 + 2d$，得 $7 = 3 + 2d$，$d = 2$，选 A。	Arithmetic sequence sum of first n terms formula: $S_n = \frac{n(a_1+a_n)}{2}$ .Given $a_3 = 7$ and $S_3 = 15$, then $15 = \frac{3(a_1+7)}{2}$, so $a_1 = 3$ .From the general term formula $a_3 = a_1 + 2d$, we get $7 = 3 + 2d$, $d = 2$, so choose A.
7	B	等差数列通项公式推论:$a_m = a_n + (m-n)d$。已知 $a_3 = 6$，$a_7 = 22$，则 $22 = 6 + 4d$，解得 $d = 4$，选 B。	Arithmetic sequence general term formula corollary: $a_m = a_n + (m-n)d$ .Given $a_3 = 6$ and $a_7 = 22$, then $22 = 6 + 4d$, so $d = 4$, choose B.
8	A	等比数列前 n 项和公式:$S_n = a_1(1 + q + q^2 + \ldots + q^{n-1})$。已知 $a_1 = 3$，$S_3 = 39$，则	Geometric sequence sum of first n terms formula: $S_n = a_1(1 + q + q^2 + \ldots + q^{n-1})$ .Given $a_1 = 3$ and

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		$3(1+q+q^2)=39$，即 $q^2+q-12=0$。解得 $q=3$ 或 $q=-4$，因数列递增且 $a_1>0$，故 $q=3$，选 A。	$S_3=39$，then $3(1+q+q^2)=39$，i.e.，$q^2+q-12=0$.Solve for q：$q=3$ or $q=-4$. Since the sequence is increasing and $a_1>0$，$q=3$，so choose A.
9	C、D	等差数列定义:后项 - 前项 = 常数(公差 d)。 - A: $(1+\sqrt{2})-1=\sqrt{2}$，$(1+\sqrt{3})-(1+\sqrt{2})=\sqrt{3}-\sqrt{2}$，不相等，非等差数列；- B:$4-1=3$，$16-4=12$，不相等，非等差数列；- C:$\lg 9-\lg 3=\lg 3$，$\lg 27-\lg 9=\lg 3$，相等，是等差数列；- D:$1-(-4)=5$，$6-1=5$，相等，是等差数列。题目若为多选题，选 C、D；若为单选题，推测题干疏漏，优先选 C。	Arithmetic sequence definition: Subsequent term - Previous term = Constant (common difference d).- A: $(1+\sqrt{2})-1=\sqrt{2}$，$(1+\sqrt{3})-(1+\sqrt{2})=\sqrt{3}-\sqrt{2}$, not equal, not an arithmetic sequence;- B: $4-1=3$，$16-4=12$，not equal, not an arithmetic sequence;- C: $\lg 9-\lg 3=\lg 3$，$\lg 27-\lg 9=\lg 3$, equal, it is an arithmetic sequence;- D: $1-(-4)=5$，$6-1=5$, equal, it is an arithmetic sequence.If the question is a multiple-choice question, choose C and D; if it is a single-choice

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			question, it is speculated that there is an oversight in the question stem, and C is preferred.
10	B	等比数列通项公式:$a_n = a_1q^{n-1}$。已知 $a_1 = 4$，$q = \frac{1}{3}$，$n = 3$，则 $a_3 = 4 \times \left(\frac{1}{3}\right)^2 = \frac{4}{9}$，选 B。	Geometric sequence general term formula: $a_n = a_1q^{n-1}$.Given $a_1 = 4$, $q = \frac{1}{3}$, $n = 3$, then $a_3 = 4 \times \left(\frac{1}{3}\right)^2 = \frac{4}{9}$, so choose B.

二、填空题（II. Fill-in-the-Blank Questions）

题号	答案	解析（Explanation）	英文解析（English Explanation）
1	$\frac{4}{7}$	由通项公式 $a_n = \frac{a_{n-1}}{2^n-1}(n$â ¥$2)$，$a_2 = 4$，则 $a_3 = \frac{a_2}{2^3-1} = \frac{4}{8-1} = \frac{4}{7}$。	From the general term formula $a_n = \frac{a_{n-1}}{2^n-1}(n$â ¥$2)$and $a_2 = 4$，then $a_3 = \frac{a_2}{2^3-1} = \frac{4}{8-1} = \frac{4}{7}$.
2	$\frac{9}{2}$	化简通项公式:$a_n = (-3)^2$Â$\cdot\frac{1}{n} = 9$Â$\cdot\frac{1}{n}$，则 $a_2 = \frac{9}{2}$。	Simplify the general term formula: $a_n = (-3)^2$Â$\cdot\frac{1}{n} = 9$Â$\cdot\frac{1}{n}$, then $a_2 = \frac{9}{2}$.
3	3；11	由递推公式 $a_n = 3a_{n-1} + 2(n > 1)$:$a_2 = 3 \times \frac{1}{3} + 2 = 1 + 2 = 3$；$a_3 = 3 \times 3 + 2 = 9 + 2 =$	From the recursive formula $a_n = 3a_{n-1} + 2(n > 1)$: $a_2 = 3 \times \frac{1}{3} + 2 = 1 + 2 = 3$； $a_3 = 3 \times 3 + 2 =$

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		11。	$9+2=11$.
4	3	数列项与前 n 项和关系：$a_n=S_n-S_{n-1}$(nâ ¥2)。$a_3=S_3-S_2=(3\times3-5)-(3\times2-5)=4-1=3$。	Relationship between sequence terms and sum of first n terms: $a_n=S_n-S_{n-1}$(nâ ¥2). $a_3=S_3-S_2=(3\times3-5)-(3\times2-5)=4-1=3$.
5	7；14；33；54	由通项公式 $a_n=n^3+6$: $a_1=1^3+6=7$；$a_2=2^3+6=14$；$a_3=3^3+6=33$；$S_3=7+14+33=54$。	From the general term formula $a_n=n^3+6$: $a_1=1^3+6=7$; $a_2=2^3+6=14$; $a_3=3^3+6=33$; $S_3=7+14+33=54$.
6	12	等差中项公式:$A=\frac{11+13}{2}=12$。	Arithmetic mean formula: $A=\frac{11+13}{2}=12$.
7	-20	等差数列通项公式:$a_7=a_1+6d=4+6\times(-4)=4-24=-20$。	Arithmetic sequence general term formula: $a_7=a_1+6d=4+6\times(-4)=4-24=-20$.
8	3	等差数列通项公式推论:$a_7=a_1+6d$，则 $20=2+6d$，解得 $d=3$。	Arithmetic sequence general term formula corollary: $a_7=a_1+6d$, then $20=2+6d$, so $d=3$.
9	$\frac{7}{2}$；4	通项公式推论:$a_5=a_1+4d$，则 $11=-3+$	General term formula corollary: $a_5=a_1+4d$, then

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		$4d$，解得 $d=\frac{7}{2}$；$a_3=a_1+2d=-3+2\times\frac{7}{2}=4$。	$11=-3+4d$, so $d=\frac{7}{2}$; $a_3=a_1+2d=-3+2\times\frac{7}{2}=4$.
10	1	通项公式:$a_7=a_1+6d$，则 $-11=a_1+6\times(-2)$，解得 $a_1=1$。	General term formula: $a_7=a_1+6d$, then $-11=a_1+6\times(-2)$, so $a_1=1$.
11	28	等比数列前 n 项和公式:$S_n=\frac{a_1(1-q^n)}{1-q}$。$S_3=\frac{4(1-(-3)^3)}{1-(-3)}=\frac{4\times28}{4}=28$。	Geometric sequence sum of first n terms formula: $S_n=\frac{a_1(1-q^n)}{1-q}$.$S_3=\frac{4(1-(-3)^3)}{1-(-3)}=\frac{4\times28}{4}=28$.
12	2	公比公式:$q=\frac{a_4}{a_3}=\frac{10}{5}=2$。	Common ratio formula: $q=\frac{a_4}{a_3}=\frac{10}{5}=2$.
13	$\hat{A}\pm2$; $\hat{A}\pm5$	通项公式推论:$a_4=a_2q^2$，则 $40=10q^2$，解得 $q=\hat{A}\pm2$；$a_1=\frac{a_2}{q}=\frac{10}{\hat{A}\pm2}=\hat{A}\pm5$。	General term formula corollary: $a_4=a_2q^2$, then $40=10q^2$, so $q=\hat{A}\pm2$; $a_1=\frac{a_2}{q}=\frac{10}{\hat{A}\pm2}=\hat{A}\pm5$.
14	1	通项公式:$a_3=a_1q^2$，则 $\frac{1}{9}=a_1\times\left(\frac{1}{3}\right)^2$，解得 $a_1=1$。	General term formula: $a_3=a_1q^2$, then $\frac{1}{9}=a_1\times\left(\frac{1}{3}\right)^2$, so $a_1=1$.
15	$\hat{A}\pm\frac{\sqrt{6}}{3}$	等比中项公式:$G=\hat{A}\pm\sqrt{\frac{\sqrt{6}}{3}\times\frac{\sqrt{6}}{3}}=\hat{A}\pm\sqrt{\frac{6}{9}}=\hat{A}\pm\frac{\sqrt{6}}{3}$。	Geometric mean formula: $G=\hat{A}\pm\sqrt{\frac{\sqrt{6}}{3}\times\frac{\sqrt{6}}{3}}=\hat{A}\pm\sqrt{\frac{6}{9}}=\hat{A}\pm\frac{\sqrt{6}}{3}$.

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16	Â ± 6	等比中项公式：$G = \text{Â} \pm \sqrt{3 \times 12} = \text{Â} \pm \sqrt{36} = \text{Â} \pm 6$。	Geometric mean formula: $G = \text{Â} \pm \sqrt{3 \times 12} = \text{Â} \pm \sqrt{36} = \text{Â} \pm 6$.
17	-10；4	通项公式:$a_8 = a_1 + 7d$，则 $11 = a_1 + 7 \times 3$，解得 $a_1 = -10$；前 n 项和:$S_8 = \frac{8(a_1+a_8)}{2} = 4 \times (-10 + 11) = 4$。	General term formula: $a_8 = a_1 + 7d$, then $11 = a_1 + 7 \times 3$, so $a_1 = -10$; Sum of first n terms: $S_8 = \frac{8(a_1+a_8)}{2} = 4 \times (-10 + 11) = 4$.
18	-4	通项公式:$a_7 = a_1 + 6d = -6 + 6 \times \frac{1}{3} = -6 + 2 = -4$。	General term formula: $a_7 = a_1 + 6d = -6 + 6 \times \frac{1}{3} = -6 + 2 = -4$.
19	1	通项公式推论:$a_8 = a_3 + 5d$，则 $15 = 5 + 5d$，解得 $d = 2$；$a_1 = a_3 - 2d = 5 - 4 = 1$。	General term formula corollary: $a_8 = a_3 + 5d$, then $15 = 5 + 5d$, so $d = 2$; $a_1 = a_3 - 2d = 5 - 4 = 1$.
20	20	先求公差:$d = \frac{a_4-a_1}{3} = \frac{8-2}{3} = 2$；前 n 项和:$S_4 = \frac{4(a_1+a_4)}{2} = 2 \times (2 + 8) = 20$。	First find the common difference: $d = \frac{a_4-a_1}{3} = \frac{8-2}{3} = 2$; Sum of first n terms: $S_4 = \frac{4(a_1+a_4)}{2} = 2 \times (2 + 8) = 20$.
21	19	通项公式推论:$a_6 = a_2 + 4d$，则 $11 = 3 + 4d$，解得 $d = 2$；$a_{10} = a_6 + 4d = 11 + 8 =$	General term formula corollary: $a_6 = a_2 + 4d$, then $11 = 3 + 4d$, so $d = 2$; $a_{10} = a_6 + 4d = 11 + 8 =$

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		19。	19.
22	3	公比公式:$q=\frac{a_4}{a_3}=\frac{15}{5}=3$。	Common ratio formula: $q=\frac{a_4}{a_3}=\frac{15}{5}=3$.
23	3	通项公式:$a_4=a_1q^3$，则 $9=\frac{1}{3}q^3$，解得 $q^3=27$，$q=3$。	General term formula: $a_4=a_1q^3$，then $9=\frac{1}{3}q^3$，so $q^3=27$，$q=3$.
24	2	公比公式:$q=\frac{a_6}{a_5}=\frac{6}{3}=2$。	Common ratio formula: $q=\frac{a_6}{a_5}=\frac{6}{3}=2$.
25	62	前 3 项和:$S_3=a_1+a_1q+a_1q^2=2+10+50=62$。	Sum of first 3 terms: $S_3=a_1+a_1q+a_1q^2=2+10+50=62$.
26	Â ± 2；2	通项公式推论:$a_5=a_3q^2$，则 $32=8q^2$，解得 $q=$ Â ± 2； $a_1=\frac{a_3}{q^2}=\frac{8}{4}=2$。	General term formula corollary: $a_5=a_3q^2$，then $32=8q^2$, so $q=$ Â ± 2； $a_1=\frac{a_3}{q^2}=\frac{8}{4}=2$.
27	8	等差中项公式:$A=\frac{7+9}{2}=8$。	Arithmetic mean formula: $A=\frac{7+9}{2}=8$.
28	$\frac{5\sqrt{7}}{2}$	等差中项公式:$A=\frac{\sqrt{7}+4\sqrt{7}}{2}=\frac{5\sqrt{7}}{2}$。	Arithmetic mean formula: $A=\frac{\sqrt{7}+4\sqrt{7}}{2}=\frac{5\sqrt{7}}{2}$.
29	$2a$	等差中项公式:$A=\frac{(2a+b)+(2a-b)}{2}=\frac{4a}{2}=2a$。	Arithmetic mean formula: $A=\frac{(2a+b)+(2a-b)}{2}=\frac{4a}{2}=2a$.

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30	$Â \pm 5\sqrt{3}$	等比中项公式：$G = Â \pm \sqrt{5 \times 15} = Â \pm \sqrt{75} = Â \pm 5\sqrt{3}$。	Geometric mean formula: $G = Â \pm \sqrt{5 \times 15} = Â \pm \sqrt{75} = Â \pm 5\sqrt{3}$.
31	$Â \pm \sqrt{2}$	等比中项公式:$G = Â \pm \sqrt{\frac{\sqrt{8}}{2} \times \frac{\sqrt{8}}{2}} = Â \pm \sqrt{\frac{8}{4}} = Â \pm \sqrt{2}$。	Geometric mean formula: $G = Â \pm \sqrt{\frac{\sqrt{8}}{2} \times \frac{\sqrt{8}}{2}} = Â \pm \sqrt{\frac{8}{4}} = Â \pm \sqrt{2}$.
32	9； $Â \pm 3\sqrt{5}$	等差中项:$A = \frac{3+15}{2} = 9$； 等比中项:$G = Â \pm \sqrt{3 \times 15} = Â \pm \sqrt{45} = Â \pm 3\sqrt{5}$。	Arithmetic mean: $A = \frac{3+15}{2} = 9$； Geometric mean: $G = Â \pm \sqrt{3 \times 15} = Â \pm \sqrt{45} = Â \pm 3\sqrt{5}$.
33	7	通项公式:$a_3 = 3^2 - 2 = 9 - 2 = 7$。	General term formula: $a_3 = 3^2 - 2 = 9 - 2 = 7$.
34	6	项与前 n 项和关系:$a_2 = S_2 - S_1 = 6 \times (2+3) - 6 \times (1+3) = 30 - 24 = 6$。	Relationship between term and sum of first n terms: $a_2 = S_2 - S_1 = 6 \times (2+3) - 6 \times (1+3) = 30 - 24 = 6$.

三、计算题（III. Calculation Questions）

1.答案（Answer）

首项 $a_1 = 3$， 前 4 项和 $S_4 = 30$。First term $a_1 = 3$, sum of the first 4 terms $S_4 = 30$.

解析 (Explanation)

1. 等差数列通项公式推论:$a_m = a_n + (m-n)d$。已知 $a_3 = 9$, $a_7 = 21$, 则:$a_7 = a_3 + 4d \rightarrow 21 = 9 + 4d \rightarrow$ 解得 $d = 3$。2. 求首项 a_1:由 $a_3 = a_1 + 2d$, 代入 $a_3 = 9$, $d = 3{:}9 = a_1 + 2 \times 3 \rightarrow$ 解得 $a_1 = 3$。3. 求前 4 项和 S_4:前 n 项和公式 $S_n = \frac{n(a_1+a_n)}{2}$, 先求 $a_4 = a_1 + 3d = 3 + 9 = 12{:}S_4 = \frac{4\times(3+12)}{2} = 2 \times 15 = 30$。

英文解析 (English Explanation)

1. Arithmetic sequence general term formula corollary: $a_m = a_n + (m-n)d$. Given $a_3 = 9$ and $a_7 = 21$, then:$a_7 = a_3 + 4d \rightarrow 21 = 9 + 4d \rightarrow$ Solve for $d = 3$.2. Find the first term a_1: From $a_3 = a_1 + 2d$, substitute $a_3 = 9$ and $d = 3$:$9 = a_1 + 2 \times 3 \rightarrow$ Solve for $a_1 = 3$.3. Find the sum of the first 4 terms S_4: Sum of first n terms formula $S_n = \frac{n(a_1+a_n)}{2}$. First find $a_4 = a_1 + 3d = 3 + 9 = 12$:$S_4 = \frac{4\times(3+12)}{2} = 2 \times 15 = 30$.

2.答案 (Answer)

首项 $a_1 = 4$, 公差 $d = 2$。First term $a_1 = 4$, common difference $d = 2$.

解析 (Explanation)

1. 列方程:- 由 $a_2 = 6$ 得:$a_1 + d = 6$(通项公式 $a_2 = a_1 + d$); - 由 $S_5 = 40$ 得:$\frac{5(a_1+a_5)}{2} = 40$(前 n 项和公式), 化简为 $a_1 + a_5 = 16$。2. 由通项公式 $a_5 = a_1 + 4d$, 代入 $a_1 + a_5 = 16{:}\ a_1 + (a_1 + 4d) = 16 \rightarrow 2a_1 + 4d = 16 \rightarrow$ 化简为 $a_1 + 2d = 8$。3. 联立方程组:$\begin{cases} a_1 + d = 6 \\ a_1 + 2d = 8 \end{cases} \rightarrow$ 两式相减得 $d = 2$, 代入 $a_1 + d = 6$ 得 $a_1 =$

4。

英文解析 (English Explanation)

1. Set up equations:- From $a_2=6$: $a_1+d=6$ (general term formula $a_2=a_1+d$); - From $S_5=40$: $\frac{5(a_1+a_5)}{2}=40$ (sum of first n terms formula), simplified to $a_1+a_5=16$.2. From the general term formula $a_5=a_1+4d$, substitute into $a_1+a_5=16$:$a_1+(a_1+4d)=16\rightarrow 2a_1+4d=16\rightarrow$ Simplified to $a_1+2d=8$.3. Solve the system of equations:$\begin{cases}a_1+d=6\\a_1+2d=8\end{cases}\rightarrow$ Subtract the first equation from the second: $d=2$, substitute into $a_1+d=6$ to get $a_1=4$.

3.答案 (Answer)

(1) $a_1=-5$, $d=6$; (2) $a_7=31$; (3) $S_4=16$。(1) $a_1=-5$, $d=6$; (2) $a_7=31$; (3) $S_4=16$。

解析 (Explanation)

(1) 求 a_1 和 d:通项公式推论 $a_6=a_3+3d$，已知 $a_3=7$，$a_6=25$: $25=7+3d\rightarrow$ 解得 $d=6$；由 $a_3=a_1+2d$: $7=a_1+2\times6\rightarrow$ 解得 $a_1=-5$。(2) 求 a_7: $a_7=a_6+d=25+6=31$。(3) 求 S_4:前 n 项和公式 $S_4=\frac{4(a_1+a_4)}{2}=2(a_1+a_4)$；$a_4=a_1+3d=-5+18=13$；$S_4=2\times(-5+13)=2\times8=16$。

英文解析 (English Explanation)

(1) Find a_1 and d:General term formula corollary $a_6=a_3+3d$, given $a_3=7$ and $a_6=25$: $25=7+3d\rightarrow$

Solve for $d=6$; From $a_3=a_1+2d$: $7=a_1+2\times 6\rightarrow$ Solve for $a_1=-5$。(2) Find a_7: $a_7=a_6+d=25+6=31$。(3) Find S_4:Sum of first n terms formula $S_4=\frac{4(a_1+a_4)}{2}=2(a_1+a_4)$; $a_4=a_1+3d=-5+18=13$; $S_4=2\times(-5+13)=2\times 8=16$。

4.答案 (Answer)

(1) $a_1=\frac{3}{8}$, $q=2$; (2) $S_5=\frac{93}{8}$。(1) $a_1=\frac{3}{8}$, $q=2$; (2) $S_5=\frac{93}{8}$。

解析 (Explanation)

(1) 求 a_1 和 q:等比数列通项公式推论 $a_7=a_4q^3$, 已知 $a_4=3$, $a_7=24$: $24=3q^3\rightarrow q^3=8\rightarrow$ 解得 $q=2$;

由 $a_4=a_1q^3$: $3=a_1\times 2^3\rightarrow$ 解得 $a_1=\frac{3}{8}$。(2) 求 S_5:前 n 项和公式 $S_n=\frac{a_1(1-q^n)}{1-q}$: $S_5=\frac{\frac{3}{8}(1-2^5)}{1-2}=\frac{\frac{3}{8}\times 31}{1}=\frac{93}{8}$。

英文解析 (English Explanation)

(1) Find a_1 and q:Geometric sequence general term formula corollary $a_7=a_4q^3$, given $a_4=3$ and $a_7=24$: $24=3q^3\rightarrow q^3=8\rightarrow$ Solve for $q=2$; From $a_4=a_1q^3$: $3=a_1\times 2^3\rightarrow$ Solve for $a_1=\frac{3}{8}$。(2) Find S_5:Sum of first n terms formula $S_n=\frac{a_1(1-q^n)}{1-q}$: $S_5=\frac{\frac{3}{8}(1-2^5)}{1-2}=\frac{\frac{3}{8}\times 31}{1}=\frac{93}{8}$。

5.答案 (Answer)

(1) $A=6$; (2) $d=-3$; (3) $S_4=18$。(1) $A=6$; (2) $d=-3$; (3) $S_4=18$。

解析 (Explanation)

(1) 求等差中项 A:等差中项公式 $A=\frac{9+3}{2}=6$。(2) 求公差 d:通项公式 $a_3=a_1+2d$，已知 $a_1=9$，$a_3=3$:$3=9+2d\rightarrow$ 解得 $d=-3$。(3) 求 S_4:前 n 项和公式 $S_4=\frac{4(a_1+a_4)}{2}=2(a_1+a_4)$；$a_4=a_1+3d=9+3\times(-3)=0$；$S_4=2\times(9+0)=18$。

英文解析 (English Explanation)

(1) Find the arithmetic mean A:Arithmetic mean formula $A=\frac{9+3}{2}=6$。(2) Find the common difference d:General term formula $a_3=a_1+2d$, given $a_1=9$ and $a_3=3$:$3=9+2d\rightarrow$ Solve for $d=-3$。(3) Find S_4:Sum of first n terms formula $S_4=\frac{4(a_1+a_4)}{2}=2(a_1+a_4)$；$a_4=a_1+3d=9+3\times(-3)=0$；$S_4=2\times(9+0)=18$。